Taoyu (Chris) Lyu

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Education

University of Pennsylvania Aug. 2026 – Expected May 2028
M.S.E. in Electrical Engineering; Focus: Computer Engineering
Rensselaer Polytechnic Institute Aug. 2022 – May 2026
B.S. in Electrical Engineering; Minor in Economics; GPA: 3.49/4.00

Experience

University of Pennsylvania, The Radway Group Sep. 2026 – Present
Research Assistant | RISC-V SoC Verification

- Developed and integrated a SystemVerilog/SVA verification environment for Chipyard-generated TileLink-UL memory RTL, separating protocol assertions, directed stimulus, DUT adaptation, and an independent reference-memory scoreboard.
- Validated a behavioral memory fixture across four configurations spanning 8–64-bit data widths and 0–7-cycle response latencies; used five fault-injection modes to check assertion and scoreboard error detection.
- Automated Cadence Xcelium regressions with Python and a GitHub Actions self-hosted runner; required injection evidence and the expected checker failure to distinguish negative-test passes from unrelated failures.

Huichuang Medical Equipment Feb. 2025 – Aug. 2025
Research and Development Intern

- Migrated schematics and multilayer PCB layouts for a TMS device from KiCad to Altium, incorporating assigned circuit revisions across control, power, and cooling-module boards.
- Proposed an I2C-controlled IC replacement for discrete circuitry and presented it in design meetings; the team adopted the proposal.
- Independently authored and archived a Chinese technical manual of approximately 30,000 characters covering circuit analysis, calibration, operating procedures, and test workflows.

Selected Projects

SRAM-Based 4-bit MAC Accelerator Fall 2026
18 nm FinFET | Cadence Virtuoso, Spectre, Pegasus, Quantus, Python

- Designed transistor-level SRAM macros, arithmetic, and a pipelined sequencer overlapping memory reads with accumulation; verified 16-element dot products in 17 cycles at a 3.2 ns period (TT, 27°C, 0.8 V).
- Sized and laid out a 6T bitcell and 16×4 array; completed DRC/LVS and parasitic extraction, measuring post-layout read SNM of 0.14 V at 0.8 V.
- Used logical-effort-guided carry-path sizing to reduce maximum observed standalone adder delay from 931 to 852 ps (8.5%); identified SRAM control ordering as the integrated clock-period limit.
- Automated Spectre waveform checks and setup/hold characterization; passed three specified full-system vectors plus reset/restart tests with two extracted SRAM arrays and checked hold windows at all 39 DFF inputs.

Two-Wide Out-of-Order Execution Core Spring 2026
SystemVerilog, ModelSim | ECSE 4780

- Implemented a 32-bit Tomasulo-style core for an ADD/MUL/LW/SW/NOP subset, with two-wide in-order dispatch, RAT-based renaming, an eight-entry ROB, and same-cycle dependency resolution between dispatch slots.
- Designed oldest-first single-CDB arbitration with pending-result retention and resource backpressure; enforced FIFO memory issue through a four-entry LSQ and in-order architectural commit.
- Built self-checking ModelSim tests for RAW/WAR/WAW dependencies, full-queue stalls, CDB contention, memory ordering, and final machine drain, checking architectural register and memory results.

RISC-V Processor Microarchitecture Design Fall 2024
SystemVerilog, ModelSim | ECSE 4770

- Implemented multicycle and five-stage processor RTL with FSM control, operand forwarding, pipeline stalls, and branch/jump flushing; verified datapath and control behavior using self-checking testbenches.

Technical Skills

Languages: SystemVerilog, Verilog, C/C++, Python, Bash, RISC-V Assembly.
Design & Verification: RTL/FSM design, SVA, scoreboards, directed regression, transistor-level CMOS/SRAM design, logical effort, timing characterization, DRC/LVS, parasitic extraction.
Tools: Virtuoso, Spectre, Xcelium, Pegasus, Quantus, ModelSim, Quartus Prime Lite, Git, Linux, Docker, GitHub Actions.